OXFORD
INTERNATIONAL
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INTERNATIONAL GCSE
Biology
9201/1

Paper 1

Mark Scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	root		1	AO1 3.1.4b 3.2.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	chloroplast		1	AO1 3.2.1b 3.1.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	by diffusion		1	AO1 3.2.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	line going up from bottom left towards top right		1	AO2 3.2.1c
	line levels off **or** peak and fall	allow tolerance of ± half a small square	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	phloem		1	AO1 3.2.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	any **two** from: • for respiration • for storage as starch / fat / oil • to make cellulose • to make amino acids **or** to make protein	ignore energy for respiration do **not** accept produce / make / create energy allow to make fatty acids	2	AO1 3.2.1d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	30 (cm^2)	allow range from 28 to 32	1	AO2 3.1.4b 6.3.2 6.3.14

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8				AO2 3.1.4b 3.2.1a 6.3.3
	22 x 0.04 = 0.88		1	
	0.88 x 2		1	
	1.76 (cm^3)	allow 1.8 (cm^3)	1	
	or			
	0.04 x 2 = 0.08 (1)	ignore rounding to 0.1		
	0.08 x 22 (1)			
	1.76 (cm^3) (1)			

Total			**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	lung		1	AO1 3.2.5a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	any **two** from: • diaphragm moves down **or** diaphragm contracts **or** diaphragm flattens • intercostal muscles contract • ribcage moves upwards (and out) • thorax increases in volume • pressure falls in lungs • lungs fill with air	ignore ribcage increases in size allow lungs increase in size allow lungs expand	2	AO1 3.2.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	1 minute		1	AO4 3.2.5b 3.2.6g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	stopwatch / stopclock	allow second hand on clock ignore timer	1	AO4 3.2.5b 3.2.6g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	state type of exercise state time of exercise	allow examples of exercise, e.g. running allow description of intensity of exercise such as exercise at 20 repetitions per min.	1 1	AO4 3.2.5b 3.2.6g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	disease / injury	allow correct named examples e.g.TB / pneumonia / cancer / asthma / smoking allow cystic fibrosis allow air pollution allow poisonous gas	1	AO1 3.2.5d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	(mechanical) ventilator	allow CPAP / nebuliser / inhaler allow oxygen mask / tube ignore iron lung	1	AO1 3.2.5d

Total			**9**	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	liver		1	AO1 3.1.3c 3.2.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	any **two** from: • temperature • (blood) glucose (levels / concentration) • ion levels	allow oxygen **or** carbon dioxide levels allow O_2 **or** CO_2 allow pH of blood / plasma	2	AO1 3.4.2e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	sweat(ing)	allow evaporation allow perspiration	1	AO1 3.4.3a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	protein		1	AO3 3.4.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	glucose (because) it is in the kidney filtrate but not in the urine	 MP2 is dependent on MP1	1 1	AO3 3.4.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.		5-6	AO1 3.2.3s
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.		3-4	AO3 3.2.3s
	Level 1: Relevant points are made. They are not logically linked.		1-2	AO3 3.2.3s
	No relevant content		0	
	Indicative content **Advantages of a kidney transplant** • a kidney transplant will last longer than treatment with dialysis alone • reduced visits to the hospital • reduced impact on lifestyle, e.g. diet, visits to hospital, time at hospital, fluid intake • toxins / waste continually removed **or** toxins / waste does not build up in the blood • no risk of infection from needle sites **Disadvantages of a kidney transplant** • need to find a suitable tissue match • may need to wait for someone else to die • long wait time health might decline • need to take immunosuppressant drugs for the rest of their life • risk of rejection • risk of infection during surgery **or** named surgical complication For **level 2** an advantage and a disadvantage of kidney transplants are given For **level 3** advantages and disadvantages of kidney transplants are given allow converse statements that clearly relate to dialysis instead of a kidney transplant			

Total			**13**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	pacemaker	allow SAN, sinoatrial node, S.A. node	1	AO1 3.2.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	valve ensure that blood flows in the correct / one direction	ignore tricuspid allow prevents the back flow (of blood from left ventricle to left atrium)	1 1	AO1 3.2.3h

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3		max 4 marks if referring to the right side of heart or if side not given		AO1 3.2.3c/h/i
	A is pulmonary vein **or** blood vessel entering heart is pulmonary vein		1	
	B is the aorta **or** blood vessel leaving the heart is the aorta		1	
	blood moves into left atrium blood moves into left ventricle	allow one mark for reference to both atrium **and** ventricle if right side or no side mentioned.	1 1	
	reference to contraction of heart muscle (of atrium or ventricle) moving the blood		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	(blood vessel) artery (reason) highest blood pressure (of the three types of vessels)	allow aorta allow shows variations in pressure (as the heart contracts and relaxes) allow as arteries take blood away from the heart MP2 is dependent on MP1	1 1	AO2 3.2.3j

Total			**10**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	(13 mm =) 1.3 (cm) 2 x 1.3 x 3.14 8.1(64) (cm)	 allow conversion from mm to cm at any point in the calculation	1 1 1	AO2 3.1.2a 6.3.14

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	any **three** from: • gall bladder • liver • oesophagus • pancreas • duodenum • small intestine • large intestine • salivary glands • anus	if none of these marks awarded, allow **1** mark for intestine(s) allow rectum	3	AO1 3.2.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	(phagocytosis) ingests / engulfs food (so) enzymes breakdown large / insoluble molecules into small / soluble molecules (so small / soluble molecules) can be absorbed	allow ingestion described ignore bacteria / viruses / pathogens ignore eat food	1 1 1	AO2 3.2.4b 3.4.7c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	(the slime mould) does not move where sticky liquid is present **or** (the slime mould) moves where there is no sticky liquid present	allow (the slime mould) does not move where it has already been allow (the slime mould) moves away from the sticky liquid ignore slime moves to the food unqualified	1	AO3 3.3.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	(so) always searches in a new area to find food **or** avoid competition with other slime mould		1	AO3 3.3.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	(light) receptors **or** light sensitive area / spot		1	AO2 3.4.2c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.7	any **one** from: • more protection / camouflage from predators • more food available • less dehydrated	allow so (mould / liquid) won't dry out	1	AO2 3.3.2 d /f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.8	meiosis		1	AO1 3.5.2g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.9	each spore has a different (mix of) chromosomes / genes / alleles / DNA		1	AO2 3.5.1a
	(and) it is random which two spore cells will join together		1	
	(so each) zygote has a different combination of chromosomes / genes / alleles / DNA from the parent		1	

Total			**17**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	(P) cell wall (Q) cytoplasm	allow (slime) capsule	1 1	AO1 3.1.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	(in animal cells) genes are in a nucleus (in bacterial cells) some genes are found in plasmids	allow bacterial cells do not have a nucleus allow the genes / DNA is free in the cytoplasm allow genes / DNA found in bacterial cells are in a single loop	1 1	AO1 3.1.1a,c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	C (opposite G) **and** T (opposite A)		1	AO3 3.5.3j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	double helix		1	AO1 3.5.3j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	(man =) **Aa or A** and **a** (woman =) **Aa or A** and **a** offspring genotypes correctly derived**:** **AA Aa Aa aa** identification of **aa** as having cystic fibrosis (correct probability) 0.25 / $^1/_4$ / 1 in 4 / 25% / 1:3	allow alternative symbols **only** if defined allow (offspring) genotypes correct for student's parental genotypes / gametes probability must match diagram do **not** accept 3:1 / 1:4	1 1 1 1 1	AO2 AO3 3.5.3g 3.5.4a 6.3.13

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.6	different base / triplet will code for a different amino acid(s) (which means) a different sequence of amino acids (is formed) (different sequence will) lead to protein / enzyme with a different shape	ignore different / non-functional protein (different sequence will) lead to protein / enzyme with a different active site	1 1 1	AO1 3.5.3 i / k

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.7	make guide molecule for faulty (CF) gene / allele (use CRISPR) enzyme to cut out faulty (CF) gene / allele replace faulty / incorrect bases with correct bases	allow in MP1 and MP2 sequence of bases for gene / allele	1 1 1	AO2 3.5.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.8	any **one** from: • could be used to enhance normal traits (i.e. height, intelligence) • creation of 'designer babies' • against beliefs of some religious groups • unknown effect on human health • lack of research	allow descriptions such as may cause mutations ignore not ethical unqualified ignore references to identity / gender	1	AO3 3.5.3 3.5.5f

Total			**18**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	$\frac{15}{100} x\ 95\,600$ 14 340 95 600 – 14340 = 81 260 **or** reduction of 15% leaves 85% 95 600 x $\frac{85}{100}$ 81 260	allow correct subtraction using students incorrectly calculated value of 15% for **1** mark	1 1 1	AO2 3.3.4 6.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	nest building for shelter climbing to obtain food/fruit **or** climbing to avoid predators	If no other mark awarded allow nest building to avoid predators for **1** mark	1 1	AO3 3.3.4c 3.3.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	flanged / older males can make calls (using throat sacs) so attract females over long distances		1	AO2 3.4.6a
	flanged / older males are more attractive to females so females will come to them		1	
	unflanged / younger males are more active / agile so they can travel long distances to find a mate **or** younger males are more active / agile so can fight (with older males) to mate with female		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4		allow converse points if clear student is describing younger males		AO2 AO3 3.2.6b 3.4.5a
	less active but still consume food / fruit		1	
	less sugar used in respiration	allow less sugar used to release energy (for movement) do **not** accept produce / make / create energy	1	
	(so more) sugar stored as lipids / fats (in cells)		1	

Total			**11**	